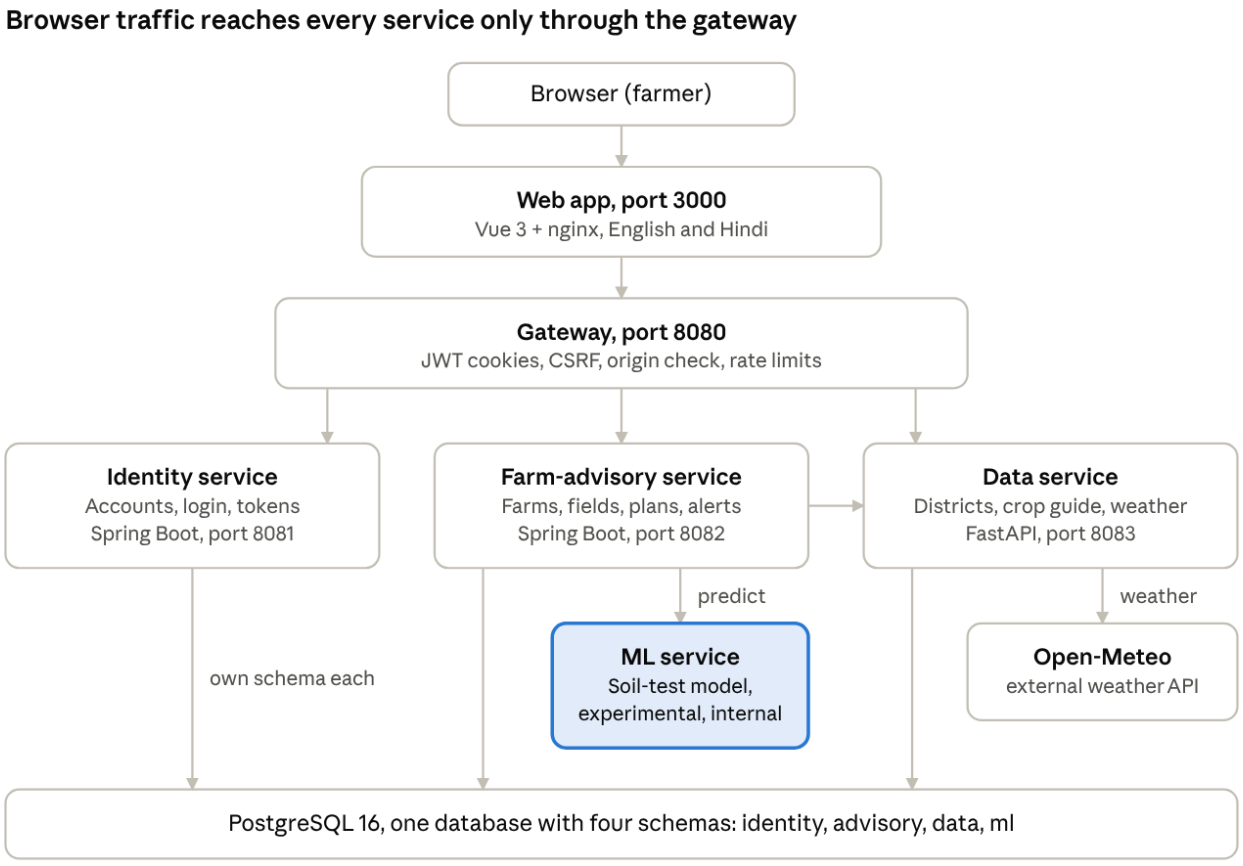
AgriSense: Architecture and Status

16 October 2026 · Jasleen Kaur

AgriSense is a farm-advisory web app: a farmer picks a district and sees which crops were historically grown there, plus weather and an experimental soil-test prediction. Everything below runs locally in Docker today; the main gaps are real-world validation, deployment and dataset licences.

Architecture



A request goes browser, web app, gateway, then one of three services. Only the farm-advisory service calls the ML service, and only the data service talks to Open-Meteo.

Status

Everything below is built, runs in Docker Compose and has automated tests. Nothing has been deployed to a server or tried on real phones.

Component	What it does	Status
Web app (Vue 3)	Landing, sign in, onboarding, crop guide, weather, prediction, schemes, alerts, admin. English and Hindi. Installable PWA.	Done, redesigned 16 Oct

Gateway	Single entry point: JWT cookies, CSRF, origin checks, rate limits	Done, 30 tests
Identity service	Register, login, refresh tokens, password reset, roles	Done, 32 tests
Farm-advisory service	Farms, fields, recommendations, action plans, alerts, feedback, prediction endpoint	Done, 45 tests
Data service	522 supported districts, Regional Crop Guide, Open-Meteo weather, nearest-district GPS lookup	Done, 32 tests
ML service	Soil-test crop model (22 crops) with SHAP factors, internal only	Done and switched on as experimental, 9 tests
Browser tests	Real-stack and mock-API suites, accessibility checks (WCAG 2 A/AA)	Done, all passing

Two features, one rule: the **Regional Crop Guide** is the main advice. It shows historical prevalence only, never a suitability score. The **soil-test prediction** is experimental, always labelled so, and never mixed into the guide.

Next

These are gaps we know about. Order is a suggestion for the team to agree on, not a committed plan.

1. **Validate the soil-test model.** Its training data looks synthetic, its rainfall period is undocumented, and its scaler had to be rebuilt. Needs real farm data or a clear "demo only" label for the final submission.
2. **Verify dataset licences.** All data sources are marked unverified in `data/SOURCES.md`. Crop records end in 2014.
3. **Hindi review.** Hindi text is a draft and needs a native speaker.
4. **Test on real phones and run CI.** Only Chromium emulation has been used, and the GitHub Actions workflow has never run.
5. **Deployment hardening.** Rate limiters live in memory (single instance only), password-reset emails are only logged (no SMTP), and the circuit breaker has no test for opening.
6. **Not planned for now:** yield prediction, fertilizer amounts, market prices, irrigation, Crop Doctor. The app shows "not available yet" pages for these rather than guessing.

Run it locally

```
cd agrisense
cp .env.example .env   # first time only
docker compose up -d --build
```

Open http://localhost:3000. Create an account on the sign-in page; the admin login is set in `.env`. Docs live in `docs/` (conventions, ADRs, ML evidence, testing record) and API contracts in `contracts/`.